

**PNGRB INVESTIGATION COMMITTEE
REPORT**

ON

MAJOR FIRE INCIDENT

AT

IOC-MATHURA REFINERY

AT

**EFFLUENT TREATMENT PLANT –
INFLUENT SUMP**

ON

7TH JANUARY 2020



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Introduction:

Indian Oil Corporation Limited has reported that there was a fire on January 07.01.2020 at around 11:47 hrs, near Influent Sump of Effluent Treatment Plant at its Mathura Refinery while the activities for replacement of spent caustic line were in progress. The fire was extinguished within 08 minute by Fire and Safety personnel of Mathura Refinery using foam tenders. Total of 6 contract workers suffered burn injuries in the incident. On 13.01.2020, one of the injured workers succumbed to his injuries in the hospital.

Indian Oil Corporation Limited conducted an internal inquiry of the incident and submitted the report. The incident investigation report of the above incident by the internal committee of IOCL, Mathura refinery is attached as **Annexure-1**.

Constitution of Enquiry Committee by PNGRB

Vide officer Order No. PNGRB/Tech/12-Inci/(1)/2019, dated 15th Jan 2020, a 3 members committee with the following persons was constituted by PNGRB to identify shortcomings, establish the cause of failure and suggest remedial measures to prevent the recurrence of such incident:

- (1) Shri S.N.Soman, GM(Operations), HPCL Mumbai
- (2) Shri R Kumar, DGM(Process Safety), BPCL Mumbai
- (3) Shri Chitrabhanu Bose, Assistant Consultant, PNGRB –Member-Convener.

A copy of the Office order Order No. PNGRB/Tech/12-Inci/(1)/2019, dated 15th Jan 2020 is attached as **Annexure-2**.

Brief description of the IOCL, Mathura Refinery Unit

Mathura Refinery, the sixth refinery of Indian Oil was commissioned in 1982 with a capacity of 6.0 MMTPA and later in the year 2000 capacity was increased to 8.0 MMTPA to meet the demand of petroleum products in north western region of the country, which includes National Capital Region. Refinery is located along the Delhi-Agra National Highway about 154 KM away from Delhi.

The major units are:

1. Crude distillation unit
2. Vacuum Distillation unit
3. Fluidized Catalytic Cracking Unit (FCCU)
4. Vis-breaker Unit (VBU)
5. Bitumen Blowing Unit (BBU)
6. Continuous Catalytic Reforming Unit (CCRU)
7. Diesel Hydro Desulphurization Unit (DHDS)
8. Once Through Hydrocracker Unit
9. Diesel Hydro-treating unit (DHDT)
10. FCC Gasoline Desulphurization (FCCGDS)

11. Selective Hydrogenation Unit (SHU)
12. Prime-G
13. Sulphur Recovery units
14. OCTAMAX unit

IOC Mathura Refinery is having its own Thermal Power Plant, which was augmented with the commissioning of three Gas Turbines (GT) and Heat Recovery Steam Generator (HRSG) in phases from 1997 to 2005 using Natural Gas (NG) as fuel to take care of environment.

Effluent Treatment Plant at IOC Mathura Refinery is designed to treat effluent at the rate of 750 m³/hr. Oil contaminated water coming out of process units, tank drains, etc. enter into Industrial Sewer Sump while contaminated rain water spillage & floor cleaning water comes into Storm Water Sewer sump.

The effluent is treated in 3 stages, namely, primary, secondary & tertiary treatment as detailed below:

- First stage of treatment includes API Oil Separator, TPI Separator & Dissolved Air Flootation (DAF) system.
- Secondary Treatment includes biological treatment of the effluent. Effluent is passed through Bio-Tower containing media followed by Aeration Tank. Effluent is then passed through clarifier to improve clarity of the effluent.
- Finally, in tertiary treatment, effluent is passed through filtration chain including Dual Media Filter & Activated Carbon Filtration to remove suspended solids and odour and routed to Guard pond.
- The waste sludge produced in the biological treatment sections are thickened in a gravity thickener. The thickened bio sludge and waste sludge produced in the oil removal sections are finally dewatered in centrifuges to produce solid sludge which is suitably disposed.
- ETP at IOC Mathura Refinery also has spent caustic treatment facility which includes treatment of spent caustic received from Process units. Sulphides present in spent caustic are removed by chemically oxidizing them to elemental Sulphur or Sulphate using strong oxidizing agent such as Hydrogen Peroxide (H₂O₂). This treated spent caustic is then processed in DAF feed sump and a portion is routed to industrial sewer sump.

Preliminary briefing of Incident back ground by IOCL, Mathura personnel:

1. IOCL Mathura Refinery was having entire Refinery shutdown (Refinery Turn around) during the above period.
2. As part of modernization job of ETP, VOC (Volatile Organic Compound) emission control system was installed on Industrial sewer Sump of ETP.

3. Since the then existing, spent caustic line to industrial sewer sump was fouling with the base frame beam of the VOC system, the above ground portion of the line was cut and removed.
4. After this, restoration of the spent caustic line to the sump by routing the line over the beam was planned as per approved P&ID.
5. For connection of the new prefabricated spool piece (size-8" & length- 3mtr) with the existing spent caustic line, the left over elbow of the existing spent caustic line above the ground near to the sump was cut and removed by gas cutting at around 11.00 hrs on 7th Jan 2020 to provide flange connection.
6. During the above job, Operations Plant In-charge took precautions of additional spark confinement of the VOC system and the job was done in his presence.
7. After this, the side cover of the VOC system located at the entry point of spent caustic line to the sump was removed for placing the prefabricated spool piece.
8. The fit up job between the prefabricated spool piece and existing spent caustic line was in progress when the Plant Manager had gone away from the site to do some other work close-by.
9. At this time, a fire broke out with a mild explosion causing injury to 4 contract workers who were working at the site and First aid injury to another 2 contract workers from the nearby area, who had come to rescue the affected contract workers.
10. The fire was extinguished within 08 minutes by the Fire & Safety of IOCL, Mathura Refinery using foam tenders after receiving the fire call at Fire station form MCP -65 & 350 at ETP area.
11. All the victims were taken to nearby Refinery Medical Center from where they were shifted to different hospitals in Mathura and Delhi based on the severity of their injuries.
12. Later, on 13.01.2020, one of the injured workers who was under treatment in Safdarjung Hospital, Delhi succumbed to his injuries.
13. Property loss (Damage of electric cables, instrument cables, Pump house, recoverable oil etc.) of estimated Rs.8 Lakhs was reported.

Details of Fatality/Injury

Total 3 contract workers of M/s. Toshiba Water solutions PVT. Ltd suffered major injuries and 1 contract worker suffered minor injury.

S.N.	Details of workers	Injury	Remarks
1	POHAP SINGH, RIGGER, 48 yrs.	80% burn injury.	Fatality (Died in the Hospital on 13th Jan 2020)
2	GIRIRAJ, RIGGER, 38 yrs.	50% burn injury	Discharged from Apollo Hospital, Delhi in February 2020
3	GEETAM SINGH, FITTER, 25 yrs.	35% burn injury	Discharged from Apollo Hospital, Delhi in February 2020..
4	OM PRAKASH, RIGGER, 32 yrs.	5% burn injury	Treatment given at IOCL, Mathura Refinery hospital and discharged.
5	ROTHAM SINGH, RIGGER		First aid given at IOCL, Mathura

			Refinery hospital and discharged on the same day
6	OM PRAKASH, HELPER		First aid given at IOCL, Mathura Refinery hospital and discharged on the same day

Terms of Reference

- Identify the lapses/shortcomings
- Establish the root causes of failure
- Suggest improvements/remedial measures to prevent recurrence of such incidents.
- Fix responsibility.

Methodology adopted for the investigation by the Committee

1. The committee visited the incident site first time on 22nd Jan 2020 & continued the investigation on 23rd Jan 2020 also.
2. The committee inspected the fire incident site and interacted with eye witnesses of IOCL Mathura & Toshiba Water Solutions Pvt. Ltd. who were involved in this activity of re-routing of existing spent caustic line to the industrial sewer sump of ETP at the site on 07.01.2020.
3. The committee also interacted with M/s. EIL (Engineers India Limited) personnel who were the coordinators for installation of VOC (Volatile Organic Compound) emission control system on Industrial sewer sump for ETP and connected piping facility.
4. Committee also screened all the related documents like log books, instructions book, medical center first hand information records, P&ID of the lines, e-mails and correspondence etc.
5. Conclusion is made based on the available site evidences, interaction with site personnel and documents.

Investigation Findings/ Observations:

Committee would like to submit that the actual site evidence could not be ascertained when first visited the site since almost 15 days had elapsed after the incident. Thus, the observations are based on the site photographs earlier taken by Internal Committee and the interaction with the concerned person present at site during the incident.

1. Incident Site Conditions:

- Total 2 no of gas detectors (GD 101 & GD 103) were provided at site on the east and north side of the Industrial Sewer sump. Gas detectors got actuated on the high readings of the %LEL indicating the presence of Hydrocarbons at the incident site. The trend of the readings is attached as **Annexure-3**.

- Level indication of the sump is available in the control room of the Effluent treatment plant. Before the incident, level of the sump was 60%. The sump pump was on line during the incident.
- There was no Public Address (PA) system available at the IS sump area for effective communication between the control room and site.
- Fire water header near to the Industrial Sewer sump area was found to be in dismantled condition.
- No Safety boards were available at the site.
- At the boundary (east side) of the Industrial Sewer sump area, excavation was carried out before the incident and the debris was found piled at the site itself.
- The structural beam was also hindering the free movement at the site. Area was found to be congested and no escape route was available.
- VOC cover was removed at the location where the fabricated spool piece for routing the spent caustic was placed.
- It was recorded that the temperature during the incident was around 16-18 deg C and visibility was clear.
- Personal gas monitor carried by one of the IOCL personnel was showing Hydro carbon presence intermittently when the committee visited the site on 22nd Jan 2020 during morning period.
- Both electrical and instrument cables, pump house shutters, FRP of VOC covering for viewing inside the sump got burnt due to the fire.
- The existing spent caustic line has 2 routings, one to Dissolved Air Filtration unit (DAF) of ETP and the other to the Industrial Sewer Sump (ISS). The existing spent caustic line to sump was isolated only through a block valve and the line to DAF was in continuous service.

2. Interaction with personnel of IOCL, Toshiba Water Solutions Pvt. Ltd & EIL:

- **ETP Manager of IOCL** Shri S K Mondal stated that the activity of restoring/rerouting of the spent caustic line to the Industrial Sewer sump was discussed with Project Sr. Manager Shri S K Singh and contractor supervisor of Toshiba Water Solutions Pvt. Ltd Shri Umakant Panda on 4th and 6th Jan 2020. The fabrication of the spool piece was completed only by around 16.30 hrs on 6th Jan 2020 which was verified jointly by ETP Manager, Project Sr. Manager and Contractor supervisor at the fabrication yard. Since the hot work permit cannot be issued after daylight hours, the activity was scheduled on 7th Jan 2020.

The left over elbow of the existing spent caustic line was removed by gas cutting in the presence of ETP Manager at around 11.00hrs on 7th Jan 2020 for fixing the prefabricated spool piece.

ETP Manager admitted that the existing spent caustic line was not positively isolated (by providing a spade downstream of the block valve) and line was not water flushed before gas cutting. Gas cutting of the pipe was carried out using the common hot work permit taken at ETP area .

Before gas cutting, VOC openings were covered with additional aluminum sheets and assured that personal gas responder reading is showing NIL presence of hydrocarbon. ETP Manager did not give permission to the contractor for doing grinding at site and contractor supervisor agreed to this.

After the job of cutting the left over elbow of the spent caustic line, ETP Manager moved to the other area of ETP for his routine round.

After a while, he heard a minor explosion and saw black smoke at the Industrial Sewer sump area and rushed to the site. He also informed fire station through the UHF set.

- **ETP - Chief Manger** Shri Phalguni Pal and **Sr.Manager** Shri R K Sharma stated that they were not aware of the replacement job of spent caustic line job on 7th January and admitted that there was no interaction with ETP Manager regarding this critical job.
- On interaction with **ETP-Operator and Shift-in -Charge**, it was observed that they were not having any knowledge regarding this job of replacement of spent caustic line and were not involved in this activity, although they were involved in issuing the permit. Though the ETP shift crew was aware of the incident, they did not have the understanding /knowledge on how the fire broke out.
- **Project Executing Engineer Mr.Anmol Walia** stated that on 7th Jan 2020 he renewed the hot work permit at ETP area which was taken on 1st Jan 2020. As per him, the permit was renewed for 2 other project jobs in the ETP area and he was not aware of the replacement of the spent caustic line at Industrial Sewer sump area and confirmed that Project Sr.Manager and Contractor Supervisor have not communicated to him regarding the above job which Project Sr.Manager and contractor supervisor also agreed. Moreover he was deputed only on that day for the ETP area jobs as the originally assigned Engineers for this area were not available on that day. He also stated that for the earlier jobs at Industrial Sewer sump area, permit was taken exclusively for Industrial Sewer sump area.
- **Project Sr.Manager** Shri S K Singh stated that though he was involved in the replacement job of the spent caustic line, he was not aware that the job would be executed on 7th Jan 2020. He admitted that Job safety plan was not prepared for this job. He agreed that he has advised Mr.Anmol to renew permits at ETP area for the project related jobs on 7th Jan 2020, as his subordinates were not available on that day.

He stated that he came to the site before the fire incident and he was discussing with the contract supervisor at that time the fire broke out with a mild explosion.

- **Contractor supervisor of Toshiba Water Solutions Pvt. Ltd** Mr. Umakant Panda stated that when the incident took place, he was at the site. He stated that as per the discussion he had with Project Sr.Manager and ETP Manager and as per their advice, he scheduled the work on 7th Jan 2020.He stated that the spool piece is fabricated on the basis of hand sketch, as there was no response from EIL even after sending them mail or the detailed sketch (the copy of the mail has not been handed over to the Committee).

He did the gas cutting of the pipe line after taking precautions as instructed by the ETP Manager. He also confirmed that ETP Manager did not give permission for grinding. After that the fit-up job between prefabricated spool piece and the existing spent caustic line was in progress after removing the VOC cover at the entry location of the spent caustic line. During this period ETP manager was not at the site. Since the flange had to be welded , contract supervisor made arrangements for doing the welding job by placing the welding machine, cable connection to the welding machine from the Electrical substation, arranging welder etc.

He told that he was discussing with Project Sr.Manager who was available at the site at that time, then suddenly the fire broke out with mild explosion. He immediately activated the Manual call point No.65 /350on the road side and rushed to site to rescue his workers. During the rescue activity he had minor fire burns on his fingers, which he treated in the IOCL Occupational health center. He stated that 4 of his workers were there at the site when fire broke out.

He also stated that since the safety officer was not available for Toshiba Water Solutions Pvt. Ltd for the past 1 month, he was acting as the safety officer. EIL agreed that they were aware of this arrangement.

- On interaction with **EIL Site In charge** Shri C. Duraiyasan, he stated initially that EIL has completed all the project jobs and is not connected with the replacement of the spent caustic line job as it is a maintenance job and no drawing was provided for this job from EIL.

Even after producing the P&ID diagram which clearly indicates the rerouting of spent caustic line, he denied having any knowledge of this rerouting of spent caustic line.

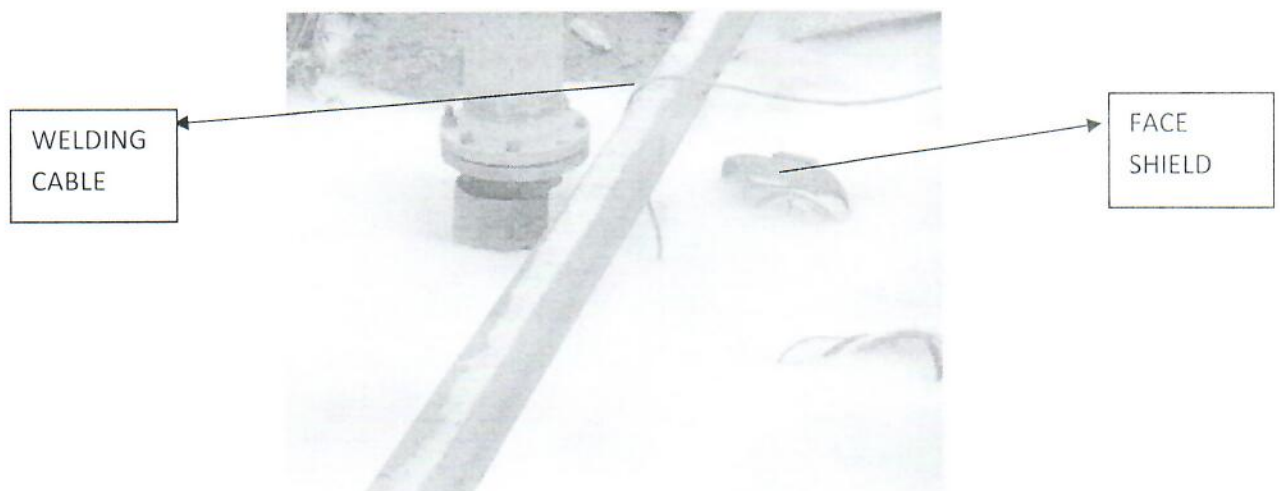
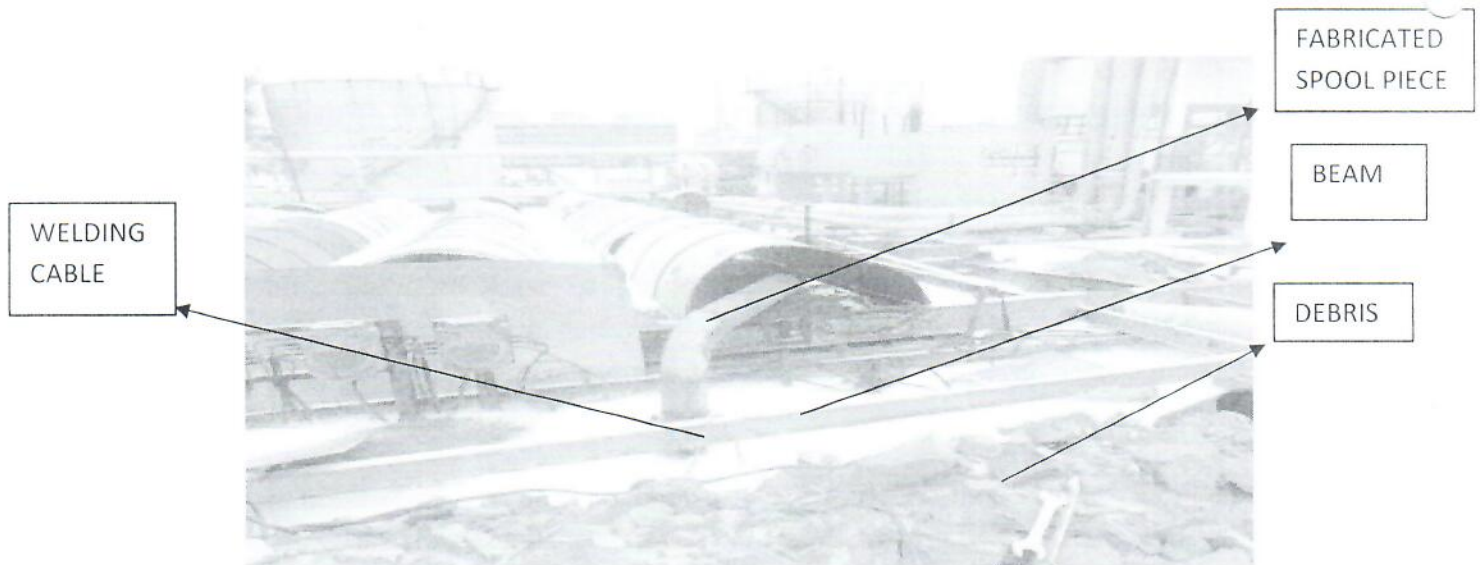
- **EIL safety officer** Shri Giriraj Singh stated that he was not aware of the job at the Industrial Sewer sump area on 7th Jan 2020,and he confirmed that this was for the other 2 jobs which he was aware.

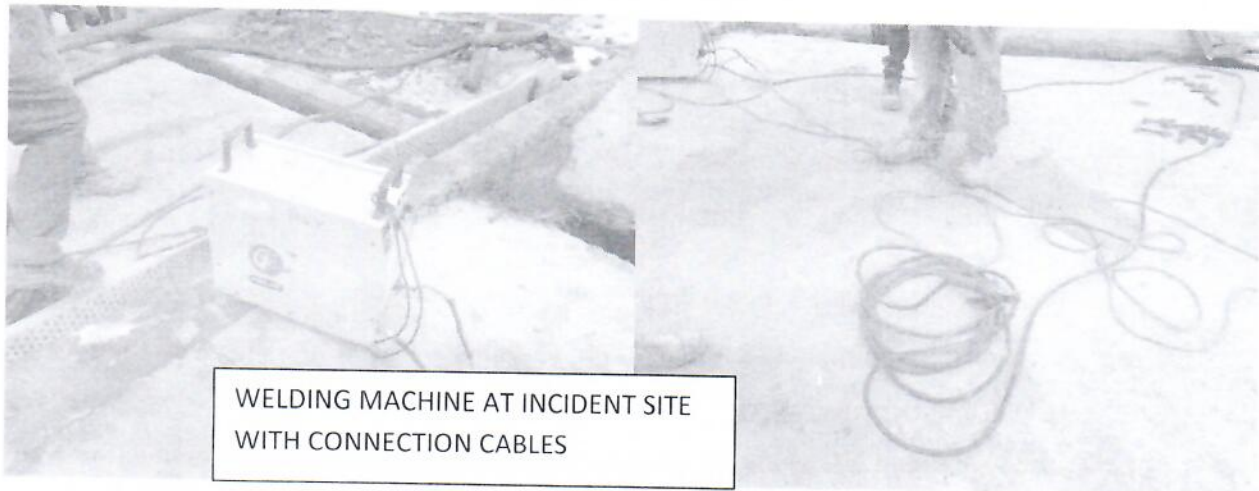
- **Inference from documents referred:**

- 11
- **As per the First Information (FIR) report** from IOCL Mathura Refinery, while carrying out the replacement job of spent caustic line to Industrial Sewer sump, a fire broke out at Industrial Sewer sump area due to ignition of Hydro carbon vapours present in the sump due to the spark generated from the incident vicinity (One of the probable source of ignition could be from Electrical cable, Instrument cable, Junction boxes, Metal contact during alignment, hot work)
 - **As per the internal incident investigation report of IOCL, Mathura Refinery**, basic cause of incident was presence of HC vapour in the IS sump and generation of spark which resulted in the fire. Root cause of the incident as per the report is improper planning and execution of the job. Gaps identified were lack of Job safety analysis, No specific work permit issued for the job at IS sump of ETP, Tool box talk was not conducted for this specific job prior to commencement of the job. It was also brought to our notice that in the internal report, there was no mention about the second fire incident which occurred on the same day at around 16.00 hrs in ETP sump area itself.
 - **Hot work permit** – The renewed hot work permit on 7th Jan 2020 for the replacement of spent caustic line job was issued on 1st Jan 2020 and the location in the permit is specified as ETP area and job is mentioned as cutting, grinding, welding etc. This permit was used for the jobs at 3 locations in ETP area and the area where gas test done was not clear as only one reading is available. As per the interaction with the personnel who were involved in the execution of this job, the communication was (only verbal) for doing this job started from 04th Jan 2020, whereas the work was carried out with a permit taken on 1st Jan 2020 which lacks clarity.
 - **Instruction book by ETP Plant manager** was found not updated after 31st Dec 2019. Hence the instruction regarding the precautions to be taken while executing the replacement of spent caustic line job was missing.
 - **Log book of ETP shift-in-charge** reveals that he is not fully involved in the job and it was not considered as critical job even after issuing the hot work permit. On referring the log book, it was also understood that there was no communication between the ETP Plant Manager and Shift incharge regarding critical jobs done in the past. Remarks column doesn't contain important information other than routine jobs.
 - **Log book of ETP operator** reveals that he was not aware of the job carried out in his area. The same operator continued for the second shift also. As per the log book for 7th Jan 2020 Day & evening shift it may be concluded that the two incidents happened on 7th Jan 2020 at around 11.47 hrs and 16.00 hrs. are mentioned in the log book at a later stage. This is confirmed from the specific observations (page 12, serial no.7) of the internal incident investigation committee report Page 12 dated 9th Jan 2020.
 - **Fire station shift in charge and Fire station Engg, Assistant log books** has mentioned both the fire incidents. The 1st fire was extinguished within 8 minutes by the Fire & Safety of IOCL, Mathura Refinery using foam tenders after receiving the

fire call at Fire station from MCP -65 & 350 at ETP area. The 2nd fire was extinguished within 5 minutes by the Fire & Safety of IOCL, Mathura Refinery using foam tenders after receiving the fire call at Fire station from ETP area.

- **Tool box talk records** indicates that the tool box talk was not conducted at the site and not related to a specific job. This was carried out in the Container and topic selected is in general.
- **Photographs** of the site incident reveal the unsafe conditions of the plant. All the debris generated during excavation was dumped at the site and there was no escape route provided. Also the beam around the sump area was obstructing the movement. From the photograph welding machine, Face shield, welding cable can be seen at the incident site.





WELDING MACHINE AT INCIDENT SITE
WITH CONNECTION CABLES



PORTION WHERE
GAS CUTTING WAS
DONE

ELBOW WHICH WAS GAS CUT AND REMOVED

- **As per the IOCL, Occupational Health Centre (OHC) report** initially 6 workers and the contract supervisor was brought to the OHC. Immediately 3 out of 6 workers were taken to nearby Refinery Medical Center from where they were shifted to different hospitals in Mathura, Delhi based on the severity of their injuries. 2 out of remaining 3 workers were given First aid at IOCL, Mathura Refinery hospital and discharged on the same day. One worker who had 5% burn injury, treatment given at IOCL, Mathura Refinery hospital and discharged after 3 days. The contract supervisor also was given First aid at IOCL, Mathura Refinery hospital and discharged on the same day.

Analysis:

Based on the above findings, observations and discussions, a detailed analysis was carried out to identify the lapses/shortcomings.

- There was no Job safety plan prepared for this critical job.
- Tool box talk for this critical job was not carried out and tool box talks were not conducted at the site.
- Communication was not at all adequate within the operation department as well as within operations, projects and executing contractor.
- There was lack of clarity regarding roles and responsibility of each personnel associated with the job.
- Effective monitoring of the job carried out by the contractor on a day to day basis by EIL was not adequate.
- Lack of coordination with EIL by the Project personnel
- Daily operating instructions were not available.

Deficiencies:

- The existing spent caustic line to sump was isolated only through a block valve and the other portion to DAF was in continuous service. Also the line was not flushed with water before handing over to Project for the rerouting job.
- Continuous monitoring of the area with portable gas monitor was not done and no communication to the panel operator to warn the ETP manager in case of higher reading of Gas detector installed at the site.
- No certified drawing was available for providing the spool piece.
- Hot work permit specific to the site work was not prepared.
- Fire water header near to the Industrial Sewer sump area was found to be in dismantled condition and no alternate arrangement was found at the site.
- Wind socks provided on the road south side of the sump was not visible from the incident site
- Area preparation of the incident site for the hot work was found to be inadequate.
- Permitting hot work when IS sump was in service and that too during Refinery shutdown (Refinery Turn around) was not reviewed properly.
- Starting the sump pump after the fire without getting clearance from the concerned Departments (In case of second Fire incident).

Conclusion:

All the findings and analysis lead to the conclusion that the incident has happened due to the following reasons:

- (a) Sufficient hydrocarbon vapor was present in and around the Sump of ETP and was continuously escaping through the openings of the sump cover. During gas cutting which took place at around 11 am, one of the side covers (at the same place where fire took place) was kept in closed position which later on was removed to fit the prefabricated spool piece (8" dia, 3 meters long). Through this side cover opening, HC gas continuously escaped and got ignited due to one of the following reasons:
 - Spark generated due to welding rod (Electrode) which might have inadvertently touched nearby metal body. The welding set fitted with wires was kept at the site and the welder was also present at the location during fire incident.
 - Spark due to cable fault or some other nearby electrical cable, Junction box could be another source of ignition/ spark generation.

Establishing the root cause(s) of failure:

- (1) Improper Job planning and improper area preparation
- (2) Not following the Safety guide lines and procedures while issuing permit and executing hot work inside the Refinery (like JSA, Tool Box meeting) has resulted in two consecutive fires within a day at a gap of 4 hrs.
- (3) Lack of awareness and competency to ascertain the critical jobs.
- (4) Lack of administrative control by Projects department over contractor job
- (5) Inter departmental and intra departmental communication gap
- (6) Not following proper SOP for executing critical jobs

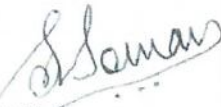
Recommendations:

- (1) Inter and Intra departmental communication on day to day basis to be improved and involvement of higher Management for periodic review of critical jobs to be ensued.
- (2) JSA and Tool box meeting to be mandatory for all for all Hot work involving Cutting/Welding/Grinding and also for critical cold work.
- (3) Regular review of the revamp/expansion jobs within running plant to be carried out by respective Division Heads of Operations with their counterpart executing agency.
- (4) Loss of Control is very serious, especially over the Contractor. Executing agency to have a strict control over the activities of their contractors and Operations to be advised not to get any job done directly from the contractor.

- (5) System of Daily Shift Instructions to Shift Operating crew to be strengthened and proper format containing all critical details to be developed and implemented.
- (6) Hierarchy Management in all the operating and executing departments must be developed and followed strictly.
- (7) A good safety culture to be developed among employees through training on Behavioral Based Safety and Process Safety to ensure that safety procedures and guide lines are not violated or bypassed.
- (8) Periodical Competency mapping of employees to ascertain the knowledge of the employees for different positions and roles and responsibilities to be carried out and training plan to be developed to bridge the gap.
- (9) Refresher training to all employees and contract workers regarding additional and specific safety measures for executing brown field expansion /revamp projects should be imparted
- (10) CCTV to be installed at critical locations. particularly in the offsite areas.
- (11) Cardinal DOs and DONTs for Process Safety to be developed and implemented.

This detailed investigation report is submitted to PNGRB as per office order issued vide No. PNGRB/Tech/12-Inci/(1)/2019, dated 15th Jan 2020.

Based on the root causes of failure of systems and procedures and the recommendations, IOC Management is requested to take a call and implement the recommendations to prevent the recurrence.


S.N. Soman
HPCL, MUMBAI


R. Kumar
BPCL, MUMBAI


Chitrabhanu Bose
PNGRB

Incident Investigation Report

RefNo.:

Date: 09.01.2020

Name of Location: Mathura Refinery	Incident No. MR/2019-20/001
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1.0 Executive Summary:

1.1	Sequence wise description of incident	- The side cover of VOC system over IS sump was removed for placing of the pre-fabricated spool. - The fit up job between pre-fabricated spool (8") of about 3 meter length and existing spent caustic line was in progress. - During alignment of the spool, the spark generated either due to metal friction or inadvertent touch of welding holder with metal which resulted into fire at 11:47 hrs.
1.2	Property loss / injury / fatality	Injury / Fatality : Injury to 04 contract workers Property Loss : Approx. Rs. 8.7 Lacs.
1.3	Cause	- HC vapour from IS Sump and generation of spark either due to metal friction or inadvertent touch of welding holder with metal during alignment of spent caustic line resulting in fire. - Improper planning and execution of the job
1.4	Recommendations	- Work permit should be taken for specific jobs and it must have clear and complete description of the work. - Job safety Analysis shall be necessarily conducted for all hot jobs. - Tool Box talk shall be given for the specific job. - Continuous monitoring of HC vapours / toxic gases shall be ensured while carrying out hot jobs in plants under operation.

Incident Investigation Report

RefNo.:

Date: 09.01.2020

2.0 Brief description of incident:

2.1	Category (Major Reportable / Minor Reportable / Other Reportable Incident)	Major Reportable Incident [Media Coverage as per CO-HSE guidelines on Incident categorization]
2.2	Date & time	07.01.2020 at 11:47 Hrs
2.3	Location	Mathura Refinery - ETP
2.4	Brief Description	A fire incident reported at ETP Industrial Sewer (IS) sump while a spool (8") was being fitted in spent caustic header injuring 04 contract workers.
2.5	Reported by	MCP 65 & MCP 350 of ETP area Contractor personnel of M/s Toshiba Water Solutions Pvt Ltd

3.0 Constitution of Investigation Committee:

3.1	Date of formation of Investigation Committee	07.01.2020
3.2	Constituted/approved by	Name: Sh. S K Awasthi Designation: ED(HSE), RHQ
3.3	Investigation Team Convenor	Name: Sh. Churaman Designation: GM (M&I), RHQ
3.4	Details of members (Name, Designation)	1. Sh. Mrityunjay Singh, DGM (T), RHQ 2. Sh. A K Raghuvanshi, CM(F&S), MR 3. Sh. A K Abhijeet, M(HSE), RHQ

(Handwritten signatures)

Incident Investigation Report

RefNo.:

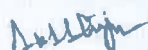
Date: 09.01.2020

4.0 Terms of Reference:

4.1	Terms of Reference of Investigation committee	• Detailed Investigation to establish the root cause of the incident. • To suggest the remedial measures to avoid such type of incident in future • To identify the technical and procedural lapses • To identify persons responsible for the incident, if any.
4.2	Methodology adopted for investigation	• Committee Meetings • Site Visits • Review of Documents • Photographs • Interviews with IOCL & contractor personnel

5.0 Brief Description of the Unit / Area / Location of Incident:

5.1	Brief Description	• Effluent Treatment Plant at Mathura Refinery is designed to treat effluent at the rate of 750 m³/hr. The effluent is treated in 3 stages, namely, primary, secondary & tertiary treatment. • Primary treatment entails removal of Oil from the Oil contaminated water. • Secondary Treatment includes biological treatment of the effluent. • Finally, in tertiary treatment, effluent is passed through filtration chain including Dual Media Filter & Activater Carbon Filtration to remove suspended solids and odour.
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Incident Investigation Report

RefNo.:

Date: 09.01.2020

6.0 Detail Description of the Incident :

6.1	Situation before the Incident	- VOC system was installed on Industrial Sewer (IS) sump. During erection of VOC system, a beam was installed at wall of the sump which was fouling with the existing spent caustic line, was removed by cold cutting to install the beam on the sump wall. - For erection of pre-fabricated spool piece, the left over elbow of old Spent caustic header was removed by gas cutting at around 11:00 Hrs on 07.01.2020. - After that, alignment / fit up job of pre-fabricated new caustic line was started.
6.2	Chronology of events leading to the Incident	- The side cover of VOC system over IS sump was removed for placing of the pre-fabricated spool. - The fit up job between pre-fabricated spool (8") of about 3 meter length and existing spent caustic line was in progress. - During alignment of the spool, the spark generated either due to metal friction or inadvertent touch of welding holder with metal which resulted into fire at 11:47 hrs.
6.3	Situation after the Incident	- Immediately after the fire incident, fire call was received at fire-station from MCP-65 & 350 of ETP area. - Fire fighting team responded immediately with fire tenders and the fire was doused within 08 mins.

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		- Contract workers (04 nos.) injured during the fire were sent to the First Aid Centre. - After that three persons were sent to Nayati Hospital for treatment while the fourth person is being treated at Mathura Refinery Hospital, two workers were released after administering first aid.
6.4	Emergency Handling / Fire Fighting Methodology adopted (in case of fire)	2 nos. Fire tenders reported at incident site. - Fire Fighting was done with application of Foam through monitors and hand lines.

7.0 Details of Fatality(s) / Injury(s):

S. No	Item Description	Fatality	Injury
7.1	Details of Person(s) injured / fatality	NII	04 contract workers of M/s Toshiba Water Solutions Pvt LTD suffered injuries: 1. Pohap Singh, Rigger, 48 Yrs, 80% burn injury 2. Giriraj, Rigger, 38 Yrs, 50% burn injury 3. Geetam Singh, Fitter, 25 Yrs, 35% burn injury 4. Om Prakash, Rigger, 32 Yrs – 5% burn injury - Further, Sh. Pohap Singh who sustained 80% has been shifted to Safdarjung Hospital, Delhi for further treatment. - Other two workers – Giriraj & Geetam Singh referred to Apollo Hospital, Delhi. - Omprakash who received 5% Injury is under treatment at MR Hospital.

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			• Other two workers Omprakash & Rotan were released after first aid at MR Hospital.
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8.0 Details of Damages / Losses:

8.1	App. Loss in case of Fire (In Rs.)	Negligible
8.2	Loss of Petroleum product due to leakage / burst (Qty) and (In Rs.)	Negligible
8.3	Any Major Disruption / Damage	No
8.4	Cost of Repair / Restoration / Maintenance (In Rs.)	Rs. 7 Lacs
8.5	Cost of Fire Fighting (In Rs.)	Rs. 1.7 Lacs
8.6	Indirect Cost (Rs.)	Under assessment
8.7	Total cost (Rs.)	Approx. Rs. 8.7 Lacs.

9.0 Methodology of Investigation:

9.1	Committee Meetings done on	07.01.2020 to 09.01.2020 at MR
9.2	Site Visits	07.01.2020 to 09.01.2020
9.3	List of Documents referred to find the facts including Police FIR / Fire Brigade Report. (Please enclose signed copies of all documents referred during investigation.)	Documents referred : 1. FIR from Mathura Refinery 2. Log book of ETP-Shift In-Charge (SIC), Log book of ETP Engg. Asst. (EA), Log book of Fire station Shift In-Charge (SIC), , Log book of Fire

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		station Engg. Asst. (EA) 3. Hot Work Permit issued on 01.01.2020 4. Record of Gas Detector reading at ETP 5. Instruction book of ETP 6. Record of Tool Box Talk of the contractor 7. Operating Manual of ETP 8. SOPs of IS Sump operation
9.4	Name of Witnesses Including eye witnesses. (Please enclose signed copies of Statements.)	1. Omprakash, Contract worker of M/s Toshiba Water Solution Pvt Ltd 2. Rotan, Contract worker of M/s Toshiba Water Solution Pvt Ltd 3. Omprakash, Contract worker of M/s Toshiba Water Solution Pvt Ltd 4. S K Mandal, PNM (ETP) 5. S K Singh, SPJM
9.5	Review of CCTV Footage, if available	CCTV facility not available for the area
9.6	Photographs of the site / Incident	Taken and enclosed as Annexure - 5
9.7	Laboratory Analysis of related materials	NA

10.0 Adequacy of Safety Systems:

10.1	Adequacy of Safety Systems	Safety systems are adequate
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11.0 Analysis of Incident:

11.1	Probable causes based on the above information (List out all the causes)	Presence of Hydrocarbon vapour at site (Level of IS sump around 60%) with the following probable source of ignition: • Spark generation from nearby sub-station • Spark generation in electrical cable and junction boxes installed nearby • Spark generation from instrument cable
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		- and junction boxes installed nearby • Spark generation through metal contact during alignment. • Hot Job in the vicinity
11.2	In case of Fire, probable source of fuel	Hydrocarbon vapours from IS Sump (Level of IS sump around 60%)
11.3	In case of Fire, probable source of ignition	• Spark generation from nearby sub-station • Spark generation in electrical cable and junction boxes installed nearby • Spark generation from instrument cable and junction boxes installed nearby • Spark generation through metal contact during alignment. • Hot Job in the vicinity
11.4	Operating Conditions: a. Power Supply b. Illumination c. Operating Parameters d. SOPs e. Remarks	a. Power Supply : 24 VDC for sump level, Gas Detector. b. Illumination : ok c. Operating Parameters : Before the incident, IS sump level was observed to be around 60-65%. d. SOPs : Available
11.5	Working Conditions a. Work Area b. Work Area Temperature c. Work Area Illumination d. Working Condition e. Inspection & Maintenance f. SOPs of inspection & Maintenance g. Remarks	a. Temperature at around 16-18°C b. Day Light c. Normal d. Maintenance activity (under project) e. SOPs for maintenance and inspection is available f. Work area contains HC environment with intermittent 5-8% LEL was observed during site visit on 08.01.2020 at 11:45 Hrs.





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11.6	Weather Conditions a. Temperature b. Rain c. Storm d. Fog e. Poor visibility f. Adequate Day Light g. Remarks	Normal winter condition. Temperature during the incident was around 16-18°C.
11.7	Analysis of all above probable causes	1. Spark generation from nearby sub-station: Not applicable as no sub-station nearby. 2. Spark generation in electrical cable and junction boxes installed nearby: As per information gathered, illumination lighting was not "ON" due to day time during the incident and pumps were also not running in vicinity. Hence, not applicable. 3. Spark generation from instrument cable and junction boxes installed nearby - As instruments installed nearby are intrinsic safe and weather proof, the possibility of spark generation from instrument cable & JB's is ruled out. 4. Spark generation through metal contact during alignment: The possibility of spark generation through metal friction cannot be fully nullified. However, the tack welding of cleats was already planned, this does not appear to be cause of fire. 5. Hot Job in the vicinity : As in this case, cleat welding job has been taken up by the welder as per his statement. However, no electrode strike mark on pipe surface was observed. It appears that the welder might have either tested welding holder or the holder got inadvertently touched resulted into generation of spark and subsequent fire.





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11.8	Inspection Record in case of equipment failure	Not Applicable
11.9	Work Instructions/ Supervision	The work was being carried out under the supervision of Plant Manager, Sh. S K Mondal, PNM and Sh. S K Singh, SPJM.

12.0 Vehicle and Driver's Details (In case of Road Accidents during transportation of petroleum products) – Not Applicable

Vehicle's Details		
S. No	Items	Details
1.	TT Number	
2.	TT Owner Flag	
3.	Chassis No.	
4.	Make	
5.	Model	
6.	Year of Manufacture	
7.	VTS Fitted (Y / N)	
8.	Remarks on VTS Fitting	
9.	ABS Fitted (Y/N)	
10.	Remarks on ABS Fitting	
11.	Speed Governor Installed (Y/N)	
12.	Remarks	
13.	Loaded (Y/ N)	
14.	Product Details (if loaded)	

Driver's Details		
S. No	Items	Details
1.	Driver Name	
2.	Age	

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3.	Driving License No	
4.	Driving License Valid Till	
5.	Received OISD Mandatory Training?	
6.	Last OISD Training received on	
7.	Holding Valid IOCL ID Card?	
8.	Drunk / Drugged?	
9.	Injured / fatality / NA	
10.	Type of Injury	
11.	Date of Hospitalization	
12.	Date of Discharge from Hospital	
13.	Medical Checkup Done?	
14.	Last Medical Checkup done on	
15.	Principle Contractor	
16.	Remarks	

13.0 Conclusion and Recommendations

13.1	Basic Cause of Incident	HC vapour from IS Sump and generation of spark either due to metal friction or inadvertent touch of welding holder with metal during alignment of spent caustic line resulting in fire.
13.2	Root cause of Incident	Improper planning and execution of the job
13.3	Gaps and Lapses If any	1. Work permit was issued for ETP Modernization work for piping erection in ETP plant area. However, no specific hot work permit was issued for replacement of Spent Caustic line to IS Sump. 2. Job Safety Analysis (JSA) was not conducted before the execution of the job. 3. Tool Box talk for this specific job was not conducted.

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		Specific Observations: 1. The communication for execution of job of replacement of spent caustic line was given on 04.01.2020. However, the work permit was issued in advance on 01.01.2020. This shows that a common work permit was issued for modernization of ETP. There was no specific permit for the mentioned job. 2. At the boundary of IS sump, excavation was done. The debris of excavation was piled at site which might have caused hindrance to the workers in escaping during the incident. 3. Time difference of 4 minutes observed between SCADA system of ETP and clock of fire control room. 4. Instruction book was not written after 31.12.2019. No specific instruction was found in the Instruction book for this job. 5. SIC Log book was not fully updated on ongoing critical activities in ETP. 6. No job description or plant abnormalities are mentioned in Operator Log book. 7. Fire Incidents occurred on 07.01.2020 was missing in operator log book.
13.4	Recommendation to avoid such types of incidents in future	Immediate : 1. Work permit should be taken for specific jobs and it must have clear and complete description of the work. 2. Job safety Analysis shall be necessarily conducted for all critical jobs. 3. Tool Box talk shall be given for the specific job. 4. Continuous monitoring of HC vapours / toxic gases shall be ensured while carrying out hot jobs in plants under operation.
		Medium Term - Nil

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		Long Term - Nil
13.5	Specific Recommendations / Remarks if any	1. Established system (Log book, Instruction book, zonal meeting, Operator shift relieving etc.) to be strictly followed. 2. All critical activities of the unit carried out during the shift should be mentioned in respective log book of SIC & area operator. 3. The filling of condition in "Additional Safety Instructions in Work Permit" should be made essential in SAP.

14.0 Incident Investigation Completion

Prepared by	By Committee members : Sh. Churaman, GM (M&I), RHQ Sh. Mrityunjay Singh, DGM (T), RHQ Sh. A K Raghuvanshi, CM(F&S), MR Sh. A K Abhijeet, M(HSE), RHQ
Reviewed By	ED & Unit Head - MR
Approved by	ED(HSE), RHQ
Enclosures	1. Annexure-1 [Office Order of Investigation] 2. Annexure -2 [Copy of FIR] 3. Annexure -3 [Copy of Hot Work Permit] 4. Annexure-4 [Statement of witnesses] 5. Annexure - 5 [Site Photographs]

1. Blower of roe system should also be commissioned simultaneously and be used.
2. Sump covers should be removed with prior permission of competent authority.
3. JSA should be re-done when the conditions are changing.
4. Critical jobs planned for day must be discussed by units heads.
5. Hot jobs should be avoid to extent possible.



पेट्रोलियम एवं प्राकृतिक गैस विनियामक बोर्ड
Petroleum and Natural Gas Regulatory Board

प्रथम-तल, वर्ल्ड ट्रेड सेंटर, बाबर रोड, नयी दिल्ली 110001 :
1st Floor, World Trade Centre, Babar Road, New Delhi – 110001
फैक्स नं. / Fax No. 91 11 23709151; E-mail contact@pngrb.gov.in

PNGRB/Tech/12-Inci/(1)/2019

15th January, 2020

Office Order

Subject: Constitution of Enquiry Committee to investigate the Fire involving Fatality at IOCL- Mathura Refinery.

Indian Oil Corporation Limited has reported that there was a fire on 07.01.2020 at around 11:47 hrs. near Influent Sump of Effluent Treatment Plant at its Mathura Refinery while the activities for replacement of spent caustic line were in progress. The fire was extinguished within 08 minutes by the fire & safety personnel of Mathura Refinery using foam tenders. 6 contract workers suffered burn injuries in the incident. On 13.01.2020, one of the injured workers succumbed to his injuries.

2. An enquiry committee comprising of the following members has been constituted to identify shortcomings, establish the cause of failure and suggest remedial measures to prevent the recurrence of such incident: -

- i. Shri S. N. Soman, GM(Operations). HPCL, Mumbai
- ii. Shri R Kumar, DGM (Process Safety), BPCL, Mumbai
- iii. Shri Chitrabhanu Bose, Assistant Consultant, PNGRB – Member - Convener

3. The terms of reference of the enquiry committee shall be as under:

- i. identify the lapses/shortcomings
- ii. establish the root cause (s) of failure
- iii. suggest improvements / remedial measures to prevent recurrence of such incidents
- iv. fix responsibility

The enquiry committee shall submit its preliminary report within 7 days and the detailed investigation report within 30 days from the date of issue of this order.

4. IOCL is advised to facilitate the investigation, provide requisite data/ records, arrange the site visits and extend necessary support to the enquiry committee.
5. The travel/stay/conveyance arrangements would be made by committee members through their respective companies / PNGRB or reimbursed on actuals.
6. Expenses related to enquiry shall be borne by PNGRB.

Vandana Sharma

(Vandana Sharma)
Secretary

Distribution

1. All Committee Members
2. Chairman, Indian Oil Corporation Limited, 3079/3 Sadiq Nagar. J B Tito Marg. New Delhi – 110 049
3. Chairman and Managing Director, Hindustan Petroleum Corporation Limited, Petroleum House, 17, Jamshedji Tata Road, Mumbai -400 020
4. Chairman & Managing Director, Bharat Petroleum Corporation Limited, Bharat Bhawan, 4&6 Currimbhoy Road, Ballard Estate, Mumbai - 400 001



Rockwell
Automation

HOME

OVERVIEW

TRENDS

ANNUNCIATOR

ALARMS

DAF-1

DAF-2

DAF-ASV

ETP-1

ETP-2

ETP-3

ETP-4

ETP-5

January 3, 2020

14:23:23

Current User: OPERATOR

ALARMS Look/View

HISTORY ALARMS



Current User: OPERATOR

PlantPA

Distributed Control Syst

Listen, Think, Solve

DETECTO

Severity

Event Time

Alarm Name

Condition Name

Message

1000

08-01-2020 12:32:48 AM

201GD_101_Aln...

TRIP

Alarm fault: Alarm input quality is bad

500

08-01-2020 12:32:48 AM

201GD_101_Aln...

TRIP

Alarm fault: Alarm input quality is bad

750

08-01-2020 12:32:48 AM

201GD_103_Aln...

TRIP

Alarm fault: Alarm input quality is bad

750

08-01-2020 12:32:48 AM

201GD_103_Aln...

TRIP

Alarm fault: Alarm input quality is bad

500

08-01-2020 12:32:48 AM

201GD_104_Aln...

TRIP

Alarm fault: Alarm input quality is bad

750

08-01-2020 12:32:48 AM

201GD_104_Aln...

TRIP

Alarm fault: Alarm input quality is bad

1000

08-01-2020 12:32:48 AM

201GD_105_Aln...

TRIP

Alarm fault: Alarm input quality is bad

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08-01-2020 12:32:48 AM

201GD_105_Aln...

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Alarm fault: Alarm input quality is bad

500

08-01-2020 12:32:48 AM

201GD_107_Aln...

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Alarm fault: Alarm input quality is bad

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Alarm fault: Alarm input quality is bad

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08-01-2020 12:32:48 AM

201GD_108_Aln...

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Alarm fault: Alarm input quality is bad

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08-01-2020 12:32:48 AM

201GD_108_Aln...

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Alarm fault: Alarm input quality is bad

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08-01-2020 12:32:48 AM

201GD_109_Aln...

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Alarm fault: Alarm input quality is bad

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Alarm fault: Alarm input quality is bad

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Alarm fault: Alarm input quality is bad

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201GD_112_Aln...

TRIP

Alarm fault: Alarm input quality is bad

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08-01-2020 12:32:48 AM

201GD_114_Aln...

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Alarm fault: Alarm input quality is bad

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08-01-2020 12:32:48 AM

201GD_114_Aln...

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Alarm fault: Alarm input quality is bad

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201GD_115_Aln...

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Alarm fault: Alarm input quality is bad

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08-01-2020 12:32:48 AM

201GD_117_Aln...

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Alarm fault: Alarm input quality is bad

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08-01-2020 05:11:35 PM

201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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07-01-2020 04:56:33 PM

201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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07-01-2020 04:56:28 PM

201GD_103_Aln...

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07-01-2020 04:50:53 PM

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Alarm fault: Alarm input quality is bad

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07-01-2020 04:50:46 PM

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Alarm fault: Alarm input quality is bad

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07-01-2020 04:49:49 PM

201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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07-01-2020 04:49:48 PM

201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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07-01-2020 12:02:48 PM

201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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07-01-2020 12:02:41 PM

201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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Alarm fault: Alarm input quality is bad

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201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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07-01-2020 11:53:35 AM

201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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07-01-2020 11:53:29 AM

201GD_103_Aln...

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Alarm fault: Alarm input quality is bad

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07-01-2020 11:53:09 AM

201GD_103_Aln...

TRIP

Alarm fault: Alarm input quality is bad

1000

07-01-2020 11:50:50 AM

201GD_103_Aln...

TRIP

Alarm fault: Alarm input quality is bad

500

07-01-2020 12:41:47 AM

201GD_103_Aln...

TRIP

Alarm fault: Alarm input quality is bad

750

07-01-2020 12:41:47 AM

201GD_117_Aln...

TRIP

Alarm fault: Alarm input quality is bad

500

07-01-2020 12:41:47 AM

201GD_117_Aln...

TRIP

Alarm fault: Alarm input quality is bad

750

07-01-2020 12:41:47 AM

201GD_117_Aln...

TRIP

Alarm fault: Alarm input quality is g

Alarm fault cleared: Alarm input quality is g

Filter: HC DETECTOR

201L_111_Aln... Fail TRIP

201GD_119_Aln... Fail TRIP

201GD_101_Aln... Fail TRIP

Advanced Analog Input Bad or Out-of-range PV: Val=24.9, Val_IncPV=24.9

Analog Input Bad PV Quality: Val=-3.5, Val_IncPV=-3.5

201GD-101 Bad PV Quality: Val=22.3, Val_IncPV=22.3